CLASS 8 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 2

Chapter 10: **Proportional Reasoning-2** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A sum of ₹ 1080 buys 30 kg of rice at ₹ 36 per kg. When the price falls to ₹ 45 per kg, the same money buys (kg): 1

(a) 30 **(b)** 22 **(c)** 60 **(d)** 24

2. A train at 80 km/h takes 6 hours for a journey. At 96 km/h it will take (hours): 1

(a) 7.2 **(b)** 6 **(c)** 4 **(d)** 5

3. Two numbers are in the ratio 5:9 and their sum is 98. Their difference is: 1

(a) 28 **(b)** 14 **(c)** 49 **(d)** 30

4. In a pie chart of 90 students a sector has a central angle of 140°. The number of students it represents is: 1

(a) 35 **(b)** 70 **(c)** 38 **(d)** 14

5. If A:B = 5:6 and B:C = 9:4, then A:C is: 1

(a) 5:3 **(b)** 5:4 **(c)** 15:8 **(d)** 20:54

6. 16 workers working 5 hours a day finish a job in 18 days. 12 workers working 10 hours a day will finish it in (days): 1

(a) 24 **(b)** 288 **(c)** 12 **(d)** 14

7. The ratio of boys to girls in a school is 9:5. If there are 1120 students, the number of girls is: 1

(a) 1106 **(b)** 400 **(c)** 720 **(d)** 560

8. A sum of ₹ 1800 buys 45 kg of rice at ₹ 40 per kg. When the price falls to ₹ 30 per kg, the same money buys (kg): 1

(a) 58 **(b)** 60 **(c)** 45 **(d)** 90

9. A train at 72 km/h takes 5 hours for a journey. At 90 km/h it will take (hours): 1

(a) 3 **(b)** 6.25 **(c)** 5 **(d)** 4

10. Two numbers are in the ratio 3:8 and their sum is 132. Their difference is: 1

(a) 11 **(b)** 60 **(c)** 62 **(d)** 66

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. A hostel has food for 60 students for 14 days. How long will the food last if 20 students leave? 2

12. A pie chart of 180 students has a sector of 72°. How many students does it represent, and what percent of the students is that? 2

13. Two numbers are in the ratio 4:11 and their sum is 225. Find the numbers. 2

14. A hostel has food for 45 students for 16 days. How long will the food last if 15 students leave? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. 20 workers working 7 hours a day can finish a job in 9 days. In how many days can 14 workers working 6 hours a day finish it? 3

16. A:B = 3:4 and B:C = 2:5. If A+B+C = 255, find the value of each. 3

17. In a pie chart, the sector for Music measures 72° and represents 20 students. Find the total number of students and the number of students represented by a 54° sector. 3

OR

24 workers working 5 hours a day can finish a job in 14 days. In how many days can 20 workers working 7 hours a day finish it? 3

18. A:B = 4:5 and B:C = 5:6. If A+B+C = 180, find the value of each. 3

OR

In a pie chart, the sector for Sports measures 80° and represents 24 students. Find the total number of students and the number of students represented by a 100° sector. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A mixture of 90 litres has milk and water in the ratio 4:5. How much water must be added to make the ratio 4:9? 5

OR

15 workers can complete a task in 24 days. After 9 days, 10 more workers join them. In how many more days will the work be finished? 5

20. A mixture of 110 litres has milk and water in the ratio 5:6. How much water must be added to make the ratio 5:7? 5

OR

20 workers can complete a task in 18 days. After 6 days, 10 more workers join them. In how many more days will the work be finished? 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND